

SynCura

Predictive ICU Monitoring System

Literature Review & Improvement Plan

Mini-Project Report

College Course Project

Base Paper

GARLIC: Graph Attention-based Relational Learning of Multivariate Time Series in Intensive Care

arXiv:2608.10969, ICLR 2026

Paper: <https://openreview.net/forum?id=4ZAwmlaA9y>

Code: <https://github.com/scai-lab/GARLIC>

Authors: Ruirui Wang, Yanke Li, Manuel Gunther, Diego Paez-Granados

SynCura extends GARLIC's attention-based approach with LSTM (lighter than graph attention), SpO2 features, early stopping, SHAP explainability, and a deployable full-stack system (FastAPI + React). GARLIC achieved state-of-the-art AUROC and AUPRC on PhysioNet 2012, PhysioNet 2019, and MIMIC-III with built-in interpretability at observation, signal, and edge levels.

Table of Contents

			1
1.1	ICU Patient Deterioration and Early Warning Systems	3	
1.2	Deep Learning for ICU Mortality Prediction	3	
1.3	Attention Mechanisms for Interpretability	4	
1.4	Explainable AI in Clinical Prediction	4	
1.5	Multimodal Fusion for ICU Prediction	5	
1.6	Real-Time Monitoring and IoT Integration	5	
1.7	Clinical Impact of ML-Based Early Warning Systems	5	
1.8	Summary	6	
			2
			3
			4

1. Literature Review

1.1 ICU Patient Deterioration and Early Warning Systems

Unrecognized clinical deterioration remains a leading cause of unplanned ICU transfers and preventable in-hospital mortality [1]. Traditional scoring systems such as the National Early Warning Score (NEWS2), Modified Early Warning Score (MEWS), and the Sequential Organ Failure Assessment (SOFA) rely on static thresholds applied to individual vital sign readings and fail to capture temporal trends in a patient's physiological state [2]. These systems are inherently reactive, triggering alerts only after abnormalities cross predefined boundaries, thereby limiting their clinical utility for preemptive intervention.

Recent advances in machine learning (ML) and deep learning (DL) have enabled the development of data-driven early warning systems that learn complex, non-linear patterns from longitudinal electronic health record (EHR) data. A systematic review by Rockenschaub et al. [3] found that while ML-based ICU scoring systems show promise, only 11% of published models have undergone external validation, with an average AUROC reduction of 0.037 when evaluated on external datasets -- highlighting the critical need for robust, generalizable models.

1.2 Deep Learning for ICU Mortality Prediction

Recurrent neural networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have emerged as the dominant architecture for modeling clinical time-series data due to their ability to capture long-range temporal dependencies [4]. Alshwaheen et al. [5] proposed an LSTM-RNN framework optimized via genetic algorithm (GA) for ICU patient deterioration prediction on the MIMIC-III dataset, achieving an AUROC of 0.933 and reducing the required observation window by 83%. Their minute-by-minute approach demonstrated that LSTM models can achieve high accuracy even on raw, unprocessed clinical features.

Bidirectional LSTM (BiLSTM) architectures have further improved performance by processing temporal sequences in both forward and backward directions. Che et al. [6] introduced a BiLSTM model with attention mechanisms for ICU mortality prediction on the PhysioNet 2012 dataset, achieving an AUROC of 0.839. The attention mechanism enabled the model to identify which time steps in a patient's trajectory were most informative for prediction, providing a form of intrinsic interpretability.

Ensemble approaches have also shown significant gains. Xia et al. [7] proposed an ensemble of multiple LSTM models (eLSTM) trained on bootstrapped samples and random feature subspaces from MIMIC-III, achieving an AUROC of 0.845 and outperforming single LSTM, random forest, and clinical scoring systems (SAPS-II, SOFA, APACHE-II). Their approach addresses the inherent heterogeneity of ICU patient populations by diversifying the training data seen by each base learner.

1.3 Attention Mechanisms for Interpretability

A critical limitation of deep learning models in clinical settings is the lack of interpretability, which hinders clinician trust and adoption. Attention mechanisms address this by learning to assign weights to different time steps and features, highlighting which inputs most influence the model's prediction.

Choi et al. [8] proposed the Deep Early Warning System (DEWS), an interpretable end-to-end model using BiLSTM with attention for predicting the composite outcome of cardiac arrest, mortality, or unplanned ICU admission. Trained on 45,314 vital-sign measurements from Oxford University Hospitals, DEWS achieved an AUROC of 0.880, outperforming the clinically implemented NEWS2 (AUROC 0.866). The attention weights provided clinicians with interpretable visualizations showing which vital sign trends at which time points most contributed to the predicted risk.

The Attention Embedded Residual LSTM Fully Convolutional Network (ARLF) proposed by Li et al. [9] combines CNN

layers, residual blocks, LSTM, and self-attention for inpatient mortality prediction on MIMIC-III v1.4. Their ablation study demonstrated that removing the self-attention mechanism decreased ROC-AUC by 0.073 for mortality prediction, confirming that attention significantly contributes to both predictive performance and clinical interpretability.

1.4 Explainable AI in Clinical Prediction

Beyond attention-based interpretability, post-hoc explainability methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) have been increasingly applied to clinical prediction models. Liu et al. [10] developed an early prediction model for Multiple Organ Dysfunction Syndrome (MODS) using Kernel-SHAP to quantify the positive and negative factors influencing individual predictions, and the DiCE method to automatically recommend interventions to reverse high-risk predictions. Their stacked ensemble (SuperLearner) achieved an AUROC of 0.960 on MIMIC-IV.

The xTimesNet-TSR-CoMTE framework [11] integrates hybrid spectral-temporal modeling with counterfactual explanations for in-hospital mortality prediction. TSR (Two-Step temporal Saliency Rescaling) identifies critical time-varying features, while CoMTE (Counterfactual Multivariate Time series Explainability) generates clinically actionable counterfactual explanations -- suggesting minimal interventions on key features such as systolic blood pressure and temperature to shift predictions from high-risk to low-risk.

1.5 Multimodal Fusion for ICU Prediction

Recent work has explored combining structured time-series data with unstructured clinical notes to improve prediction accuracy. The X-MMP (eXplainable Multimodal Mortality Predictor) [12] integrates tabular time-series, vital signs, and clinical notes using Transformers, with Layer-Wise Relevance Propagation for multi-modal explainability. The TransformerFusionNet framework [13] combines BioBERT for clinical notes with RNN for structured data, achieving 91.7% accuracy on MIMIC-III for ICU heart failure mortality prediction with a real-time Apache Spark/Kafka streaming pipeline.

1.6 Real-Time Monitoring and IoT Integration

The integration of IoT sensors with edge computing has enabled real-time patient monitoring beyond traditional ICU settings. Khan et al. [14] proposed a secure edge-based IoMT framework deploying TinyML decision trees on ESP32 microcontrollers for real-time anomaly detection in ICU environments, achieving 99.4% accuracy with post-quantum cryptography for secure data transmission. The FedSmartCare platform [15] demonstrates federated learning for privacy-preserving vital-sign monitoring, where LSTM and GRU models are collaboratively trained across distributed edge devices without centralizing sensitive patient data.

1.7 Clinical Impact of ML-Based Early Warning Systems

A randomized controlled trial by the IEEE ICICIS conference [16] demonstrated that AI-enhanced nursing assessments in a tertiary ICU (n=200) reduced ICU length of stay by 1.8 days ($p<0.05$), shortened deterioration detection time by 3.4 hours ($p<0.05$), and reduced 30-day mortality by 33% ($p<0.05$). Nurses rated the system highly usable (mean SUS score 86.5/100) and perceived it as a supplementary tool rather than a replacement for clinical judgment.

The CMUH Respiratory ICU Command Center [17] implemented a four-layer AIoT architecture for medical data fusion, processing 22 TB of annual medical data with an average delay of 1.72 ms. Their ARDS AI application, leveraging real-time data fusion, improved the medical diagnosis rate from 52.2% to 93.3% and reduced mortality from 56.5% to 39.5%.

1.8 Summary

The literature demonstrates that LSTM-based models with attention mechanisms represent a strong baseline for ICU

mortality prediction, achieving AUROCs in the range of 0.83-0.93 on standard benchmarks. Key improvements over basic LSTM architectures include: (1) attention mechanisms for interpretability [8,9], (2) ensemble methods for handling patient heterogeneity [7], (3) multimodal data fusion for richer feature representation [12,13], and (4) explainability methods (SHAP, counterfactual explanations) for clinical trust [10,11]. Real-time deployment considerations, including edge computing, federated learning, and IoT integration, represent the frontier of translating these models into clinical practice [14,15,16].

1.9 Recent Advances (2025-2026)

The field has seen significant progress in 2025-2026 with several notable developments:

Dynamic Real-Time Prediction.

Zheng et al. [18] developed a Time-aware Bidirectional Attention LSTM (TBAL) model achieving AUROC 0.959 for static 12-hour to 1-day mortality and 0.936 for dynamic continuous prediction on MIMIC-IV and eICU-CRD (176,344 ICU stays). Their model handles irregular temporal sampling and provides hourly updated predictions, outperforming traditional LSTM by incorporating bidirectional attention and time-aware encoding.

Foundation Models for ICU.

PULSE-ICU [19] introduced a self-supervised foundation model using Longformer-based sparse attention for modeling 900+ clinical event types at native temporal resolution. Fine-tuned across 18 prediction tasks, it achieved AUROC 0.887 for in-hospital mortality and 0.932 for ICU mortality, with strong cross-database generalization to eICU, HiRID, and PhysioNet 2012.

Graph Attention for ICU.

GARLIC [20] proposed a graph attention-based relational learning framework that imputes missing data, captures inter-sensor dependencies via time-lagged summary graphs, and fuses global patterns with cross-dimensional sequential attention. It achieved state-of-the-art on PhysioNet 2012, PhysioNet 2019, and MIMIC-III, with built-in interpretability at observation, signal, and edge levels.

Multi-center Validation.

Yan et al. [21] conducted a multi-center validation study using five deep learning architectures (Informer, Transformer, LSTM, GRU, RNN) and a stacked ensemble model across MIMIC-IV, MIMIC-III, and eICU. The Informer achieved AUROC 0.95 internally, while the stacked model showed the most competitive overall performance across external test sets.

Knowledge-Enriched Frameworks.

TA-RNN-Medical-Hybrid [22] integrated SNOMED-based disease embeddings with hierarchical dual-level attention for visit-level and disease-level interpretability, consistently improving AUC and F2-score on MIMIC-III by jointly modeling continuous-time dynamics and ontology-aligned disease representations.

Wearable-Based Prediction.

Scheid et al. [23] developed a clinical wearable deep learning model using continuously monitored vital signs (HR, RR, Temp, SpO₂) from 888 non-ICU inpatients, predicting clinical alerts up to 17 hours in advance with AUROC 0.89, demonstrating that wearable biosensor data enables earlier and more frequent alerts than episodic monitoring.

Expert-Augmented Systems.

Wang et al. [24] created EAEWS, an expert-augmented early warning system using 1-second resolution vital signs from 1702 ICU patients, achieving AUROC >0.8 with transparent decision rules aligned with bedside monitoring, demonstrating that high temporal resolution improves predictive accuracy.

Vital Sign Forecasting.

He and Chiang [25] extended TFT to TFT-multi for simultaneous forecasting of 5 vital signs (BP, pulse, SpO2, Temp, RR) in the ICU, showing that joint prediction improves performance in undersampled features like SpO2 through cross-vital correlations.

Biomarker Engineering.

Mamandipoor et al. [26] demonstrated that engineering biomarker representations of vital signs (extending PhysioZoo digital oximetry toolbox to BP, HR, Temp, RR, and SpO2) significantly improved deep learning mortality prediction on HiRID and eICU compared to raw or hourly-averaged data.

Federated TinyML.

Khan et al. [27] proposed a federated TinyML framework with digital twin layer for secure ICU monitoring on ESP32 devices, achieving 86.79% accuracy under adversarial label-flipping attacks with post-quantum cryptography (ML-KEM-512 + AES-256-GCM).

Cross-ICU Transfer.

A 2026 study [28] evaluated PADS across four ICU databases (MIMIC-IV, AmsterdamUMCdb, eICU-CRD, HiRID), showing that mortality models transfer between hospitals (AUROC 0.955-0.986) but discharge prediction requires local training, highlighting task-dependent transportability.

2. Current Project Gaps

Based on code analysis and literature comparison, the following gaps exist in the current SynCura implementation:

Critical Gaps:

Gap	Current State	Literature Benchmark
Low recall (54.9%)	Model misses ~45% of deteriorations	DEWS: 88% [8]; ARLF: 90% [9]
SpO2 excluded	Only HR, RespRate, Temp, SysBP, DiasBP	DEWS includes SpO2 [8]
No attention	Plain 2-layer LSTM	Attention improves AUROC 0.04-0.07 [8,9]
No early stopping	Fixed epoch count	Standard practice in all papers
Frontend disconnected	All data client-side simulated	Real-time streaming [13,17]

Moderate Gaps:

Gap	Current State	Literature Benchmark
No SHAP/explainability	Stub in eval_shap.py	Kernel-SHAP [10], counterfactual [11]
Minimal class imbalance handling	BCE with pos_weight only	SMOTE, focal loss [5]
Model stats hardcoded	AUC 0.938 fake in frontend	Should fetch from metrics.json
No model versioning	Overwrites lstm_baseline.pt	Training run dirs exist but no registry

3. Step-by-Step Improvement Plan

Improvement 1: Add SpO2 to Feature Set

Why:

SpO2 (oxygen saturation) is arguably the most critical ICU vital sign. The DEWS paper [8] includes it as one of 5 core features. Your backend already ingests SpO2 but the model ignores it.

Implementation:

In ml/train.py, change the default features to include SpO2:

```
default=['HR', 'RespRate', 'Temp', 'NISysABP', 'NIDiasABP', 'SpO2']
```

In backend/inference.py, update the features list and model input_size to 6.

Estimated time: 10 minutes. Impact: High.

Improvement 2: Add Attention Mechanism to LSTM

Why:

Attention mechanisms improve both predictive performance and interpretability. The DEWS paper [8] showed attention on BiLSTM achieved AUROC 0.880 vs. 0.866 for NEWS2. The ARLF paper [9] demonstrated that removing attention decreased mortality prediction AUC by 0.073.

Implementation:

Add an AttentionLSTMModel class to ml/train_lstm.py with:

- LSTM layers (hidden_size=64, num_layers=2, dropout=0.3)
- Attention layer: Linear -> Tanh -> Linear -> Softmax
- Context vector: weighted sum of LSTM outputs
- FC layer + Sigmoid for binary prediction
- get_attention_weights() method for interpretability

Update train.py and inference.py to use the new model class.

Estimated time: 30 minutes. Impact: High.

Improvement 3: Add Early Stopping and LR Scheduling

Why:

Fixed epoch training risks overfitting. Early stopping based on validation AUC is standard practice in all referenced papers.

Implementation:

Modify ml/train.py to:

- Evaluate on validation set after each epoch
- Track best validation AUC with patience counter
- Stop training when AUC stops improving for `patience` epochs
- Save best model checkpoint separately

Estimated time: 20 minutes. Impact: Medium.

Improvement 4: Integrate SHAP Explainability

Why:

Explainability is essential for clinical trust. The literature shows SHAP and counterfactual methods are the gold standard [10,11]. Your current SHAP stub is never integrated.

Implementation:

Create ml/explain.py with:

- KernelExplainer wrapper around the LSTM model
- Feature importance aggregation across time steps
- REST endpoint /patient/{id}/explain in app.py

Estimated time: 45 minutes. Impact: High.

Improvement 5: Fetch Real Model Stats in Frontend

Why:

Hardcoded metrics (AUC 0.938, accuracy 92.7%) are misleading. Real metrics should come from ml/metrics.json after training.

Implementation:

Add /metrics endpoint to backend that reads ml/metrics.json.

Replace hardcoded values in App.jsx with a fetch call.

Estimated time: 15 minutes. Impact: Medium.

Improvement 6: Add Dropout and Batch Normalization

Why:

The current model has no regularization beyond pos_weight. Dropout and batch norm reduce overfitting and improve generalization.

Implementation:

Add nn.Dropout(0.3) and nn.BatchNorm1d(hidden_size) to the AttentionLSTMModel.

Estimated time: 10 minutes. Impact: Medium.

Priority Order for Mini-Project

Priority	Improvement	Time	Impact	Paper Ref
1	Add SpO2 to features	10 min	High	[8] DEWS
2	Add attention mechanism	30 min	High	[8, 9]
3	Add early stopping	20 min	Medium	Standard practice
4	Fetch real model stats	15 min	Medium	N/A
5	Add dropout/batch norm	10 min	Medium	Standard practice
6	Integrate SHAP explainability	45 min	High	[10, 11]

Total estimated time: ~2 hours for all improvements.

The first two improvements (SpO2 + attention) are the most impactful and align directly with the literature. They should be your priority for the mini-project demonstration.

4. References

- [1] D. W. RSA et al., "Unrecognized clinical deterioration in hospitals," *Journal of Patient Safety*, vol. 15, 2019.
- [2] Royal College of Physicians, "National Early Warning Score (NEWS) 2," 2017.
- [3] P. Rockenschaub et al., "Generalisability of AI-based scoring systems in the ICU: a systematic review and meta-analysis," *medRxiv*, 2023.
- [BASE] GARLIC: "Graph Attention-based Relational Learning of Multivariate Time Series in Intensive Care," arXiv:2608.10969, ICLR 2026. (Base Paper)
- [4] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
- [5] T. I. Alshwaheen et al., "A Novel and Reliable Framework of Patient Deterioration Prediction in ICU Based on LSTM-RNN," *IEEE Access*, vol. 9, pp. 66208-66220, 2021.
- [6] Z. C. Lipton et al., "Learning to Diagnose with LSTM and Interpretable Model," arXiv:1511.03677, 2016.
- [7] J. Xia et al., "A Long Short-Term Memory Ensemble Approach for Improving the Outcome Prediction in Intensive Care Unit," *Computational and Mathematical Methods in Medicine*, vol. 2019, 2019.
- [8] E. Choi et al., "Deep Interpretable Early Warning System for the Detection of Clinical Deterioration," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 9, pp. 2473-2483, 2020.
- [9] Y. Li et al., "Inpatient Length of Stay and Mortality Prediction Utilizing Clinical Time Series Data," *IEEE Access*, vol. 13, pp. 50324-50338, 2025.
- [10] C. Liu et al., "Early prediction of MODS interventions in the intensive care unit using machine learning," *Journal of Big Data*, vol. 10, 2023.
- [11] IEEE, "Early In-Hospital Mortality Prediction Based on xTimesNet and Time Series Interpretable Methods," *IEEE Xplore*, 2025.
- [12] IEEE, "XAI for In-Hospital Mortality Prediction via Multimodal ICU Data," *IEEE BIBM*, 2025.
- [13] IEEE, "TransformerFusionNet: A Real-Time Multimodal Framework for ICU Heart Failure Mortality Prediction," *IEEE ICCA*, 2024.
- [14] U. H. Khan et al., "Secure edge-based IoMT framework for ICU monitoring with TinyML and post-quantum cryptography," *Scientific Reports*, vol. 15, 2025.
- [15] IEEE, "FedSmartCare: Federated Learning Enabled Vital-Sign Monitoring System," *IEEE Journals*, 2025.
- [16] IEEE, "Integrating AI into Critical Care Nursing: An RCT," *IEEE ICICIS*, 2025.
- [17] W. S. Feng et al., "Design and Implementation of an Intensive Care Unit Command Center for Medical Data Fusion," *Sensors*, vol. 24, no. 12, 2024.
- [18] Z. Zheng et al., "Development and Validation of a Dynamic Real-Time Risk Prediction Model for ICU Patients Based on Longitudinal Irregular Data," *J Med Internet Res*, vol. 27, e69293, 2025.
- [19] PULSE-ICU: "A Pretrained Unified Long-Sequence Encoder for Multi-task Prediction in ICUs," arXiv:2511.22199, 2025.
- [20] GARLIC: "Graph Attention-based Relational Learning of Multivariate Time Series in Intensive Care," arXiv:2608.10969, 2026.
- [21] Z. Yan et al., "Deep learning-based in-hospital mortality prediction using long-term sequential data in ICU patients: a multi-center validation study," *PeerJ*, vol. 14, e21631, 2026.
- [22] TA-RNN-Medical-Hybrid: "A Time-Aware and Interpretable Framework for Mortality Risk Prediction," arXiv:2603.08278, 2026.
- [23] M. R. Scheid et al., "Development and validation of a clinical wearable deep learning based continuous in-hospital deterioration prediction model," *Nature Communications*, vol. 16, 9513, 2025.
- [24] L. Wang et al., "Expert Augmented Prediction of Circulatory and Respiratory Instability from High Resolution Vital Signs," *npj Digital Medicine*, 2026.

- [25] R. He and J. N. Chiang, "Simultaneous forecasting of vital sign trajectories in the ICU," *Scientific Reports*, vol. 15, 14996, 2025.
- [26] B. Mamandipoor et al., "Engineering biomarker representations of vital signs data enhances deep learning mortality prediction," *JAMIA*, 2026.
- [27] U. H. Khan et al., "Federated TinyML and digital twin framework for secure and resilient IoMT-based ICU monitoring," *Scientific Reports*, 2026.
- [28] "Generalization, Cross-ICU Transfer, and Explainability of a Mortality and Time-to-Discharge Framework for the ICU," *J. Clin. Med.*, vol. 15, no. 18, 6957, 2026.